Introduction to R and Python for Data Science

A Student Guide for Grading, Coding Style, and Project Milestones

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Preface

This resource is intended for students in *Introduction to R and Python for Data Science*. It provides materials and guidance to support understanding the grading rubrics, coding style, and project milestones.

Introduction

This resource is intended for students in *Introduction to R and Python for Data Science*. It provides materials and guidance to support success throughout the course, including information about grading rubrics, coding style expectations, assignment requirements, and project milestones.

The guide is designed to help students understand how coursework will be evaluated and how to develop clear, readable, and well-documented code in both R and Python. It also outlines expectations for visualizations, written interpretations, and project milestones.

R Project Folder

Setting up Your RProject Folder

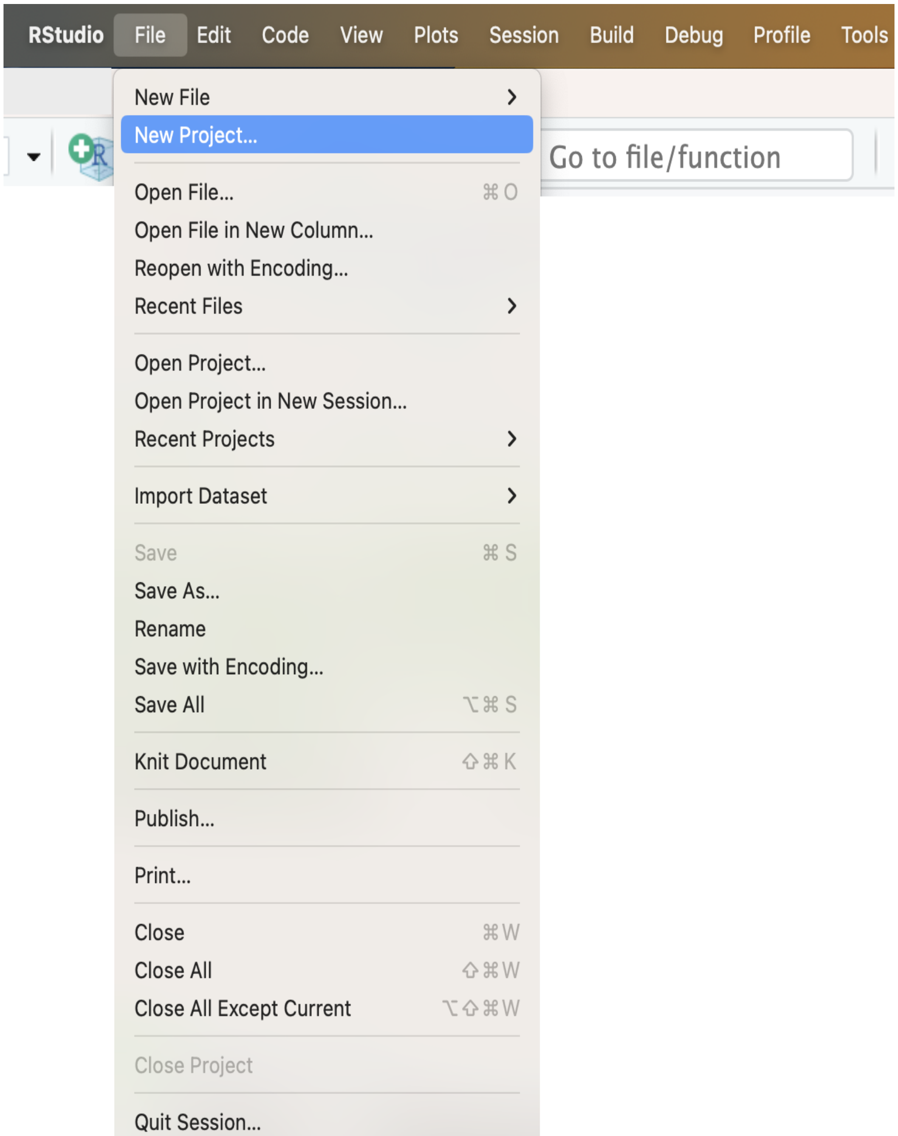
Follow these steps to:

- Create an Rproject folder named `dsc201`.
- Create a data folder inside the RProject folder.
- Download a dataset the `us_farmers_markets.csv` into the data folder.
- Download the `ds201_001-002_a00.Rmd` RMarkdown file into the RProject folder.

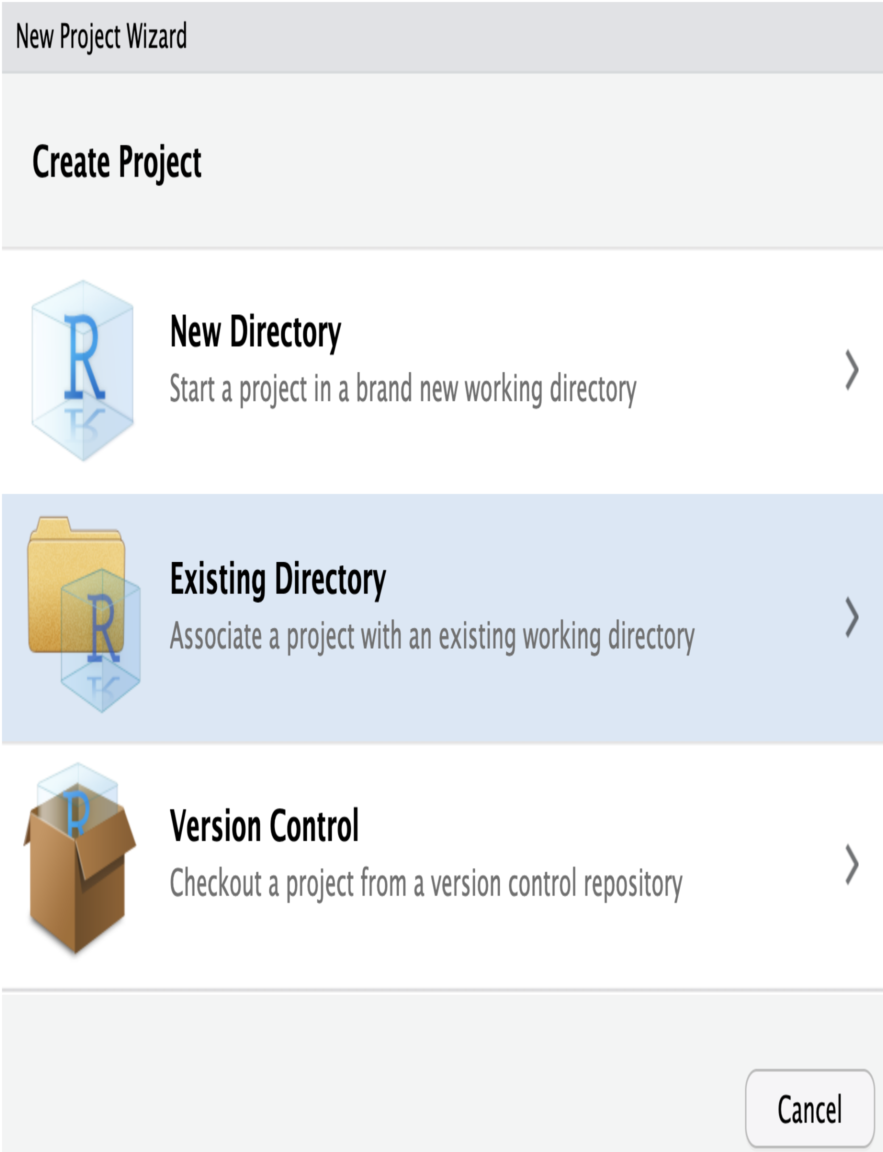
Note: Following these steps helps avoid file path errors and follows best practices from Martin Chan's RStudio Projects and Working Directories: A Beginner's Guide.

Step 1: Create a new RStudio project

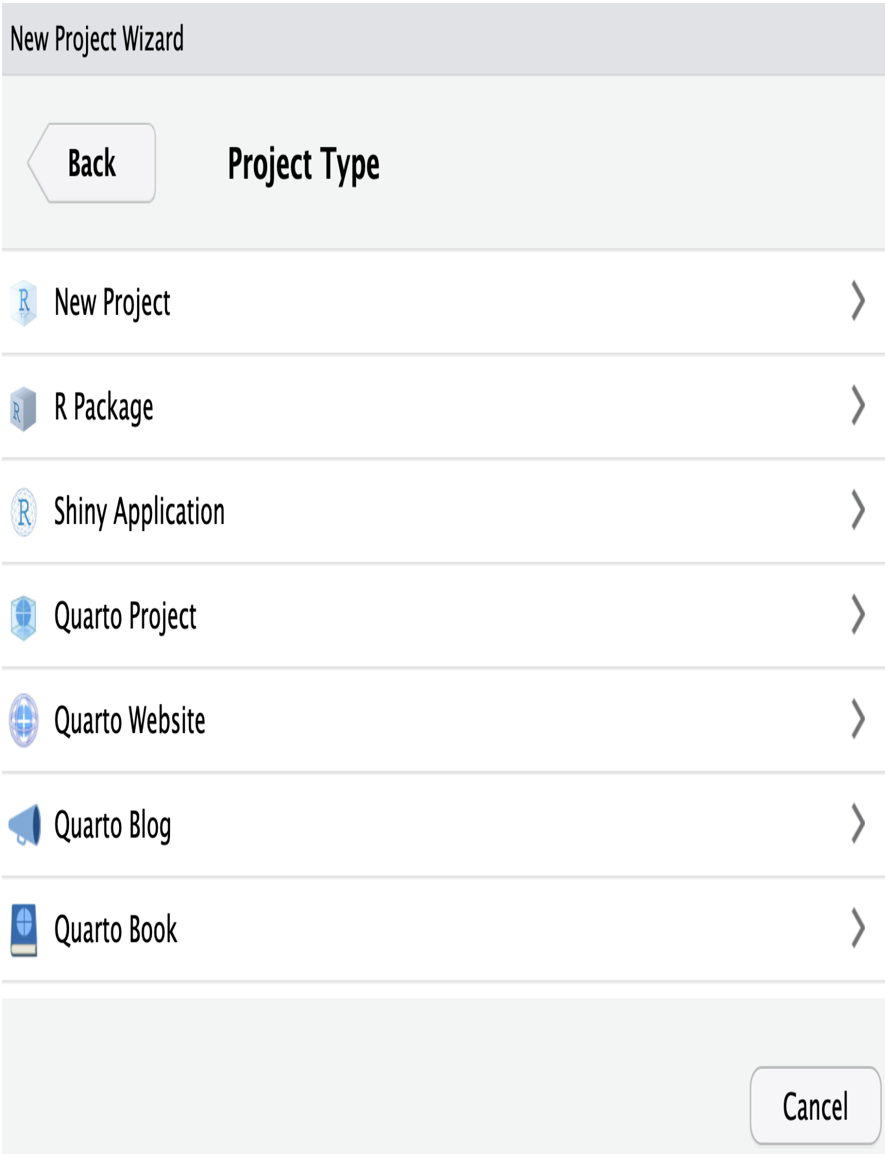
1. Open RStudio.
2. Go to **File** → **New Project...**



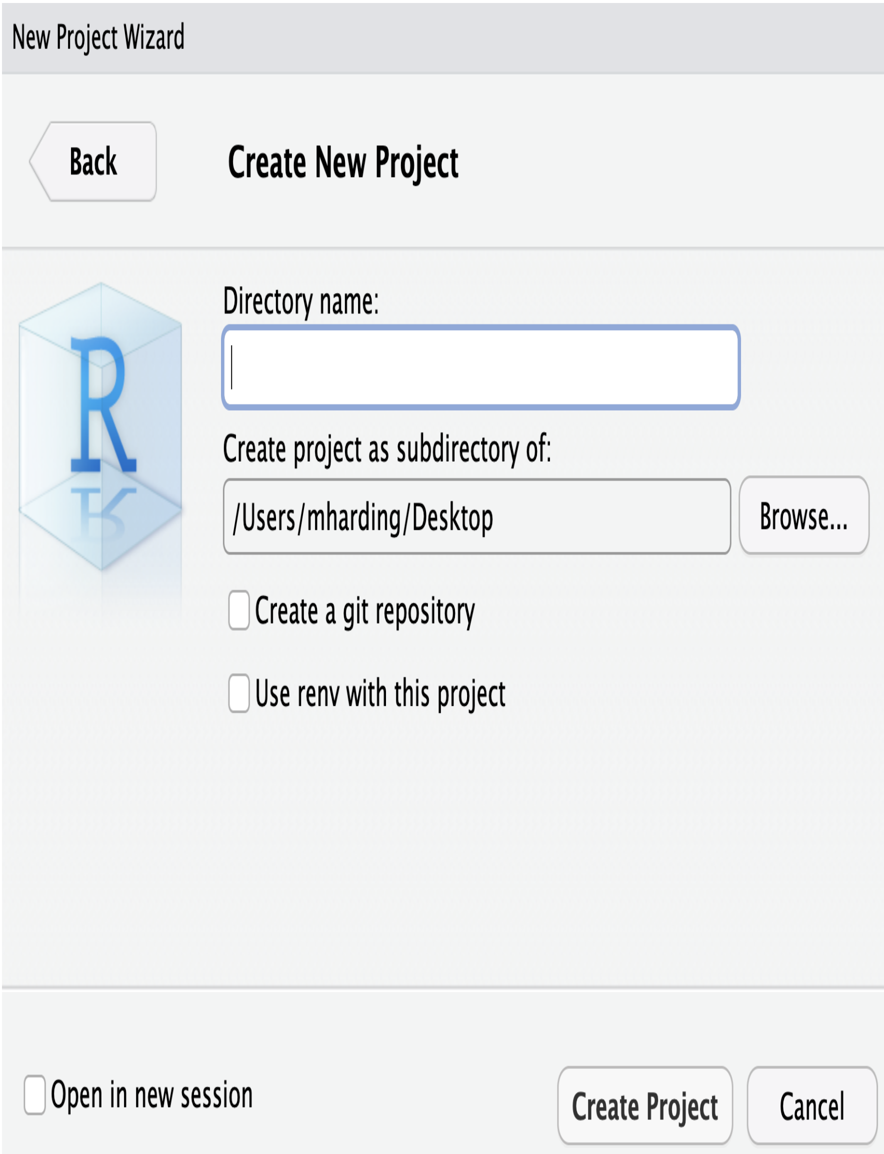
3. Select **New Directory** $\rightarrow$ **New Project**.



4. **Select New Project.**



5. In **Directory name:**, enter dsc201 (all lowercase)
6. In Create project as subdirectory of:, select Browse, then choose a location on your computer (e.g., Documents, Desktop, etc).



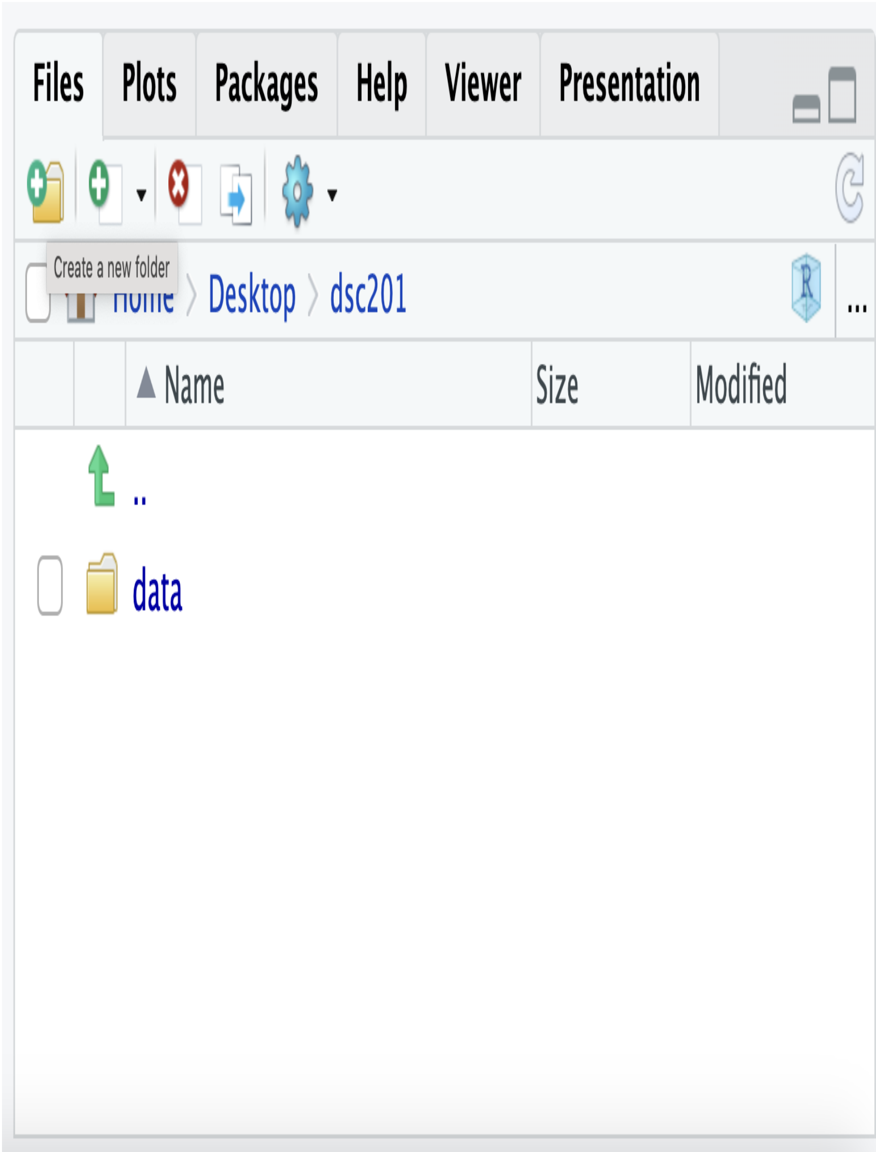
7. Click **Create Project**.

Note: Opening the `dsc201.Rproj` file will automatically set your working directory to the project's root folder.

Step 2: Create the data folder

Important: Name the folder `data` **(all lowercase)**.

8. In RStudio's **Files** pane (bottom right), click the **New Folder** icon.



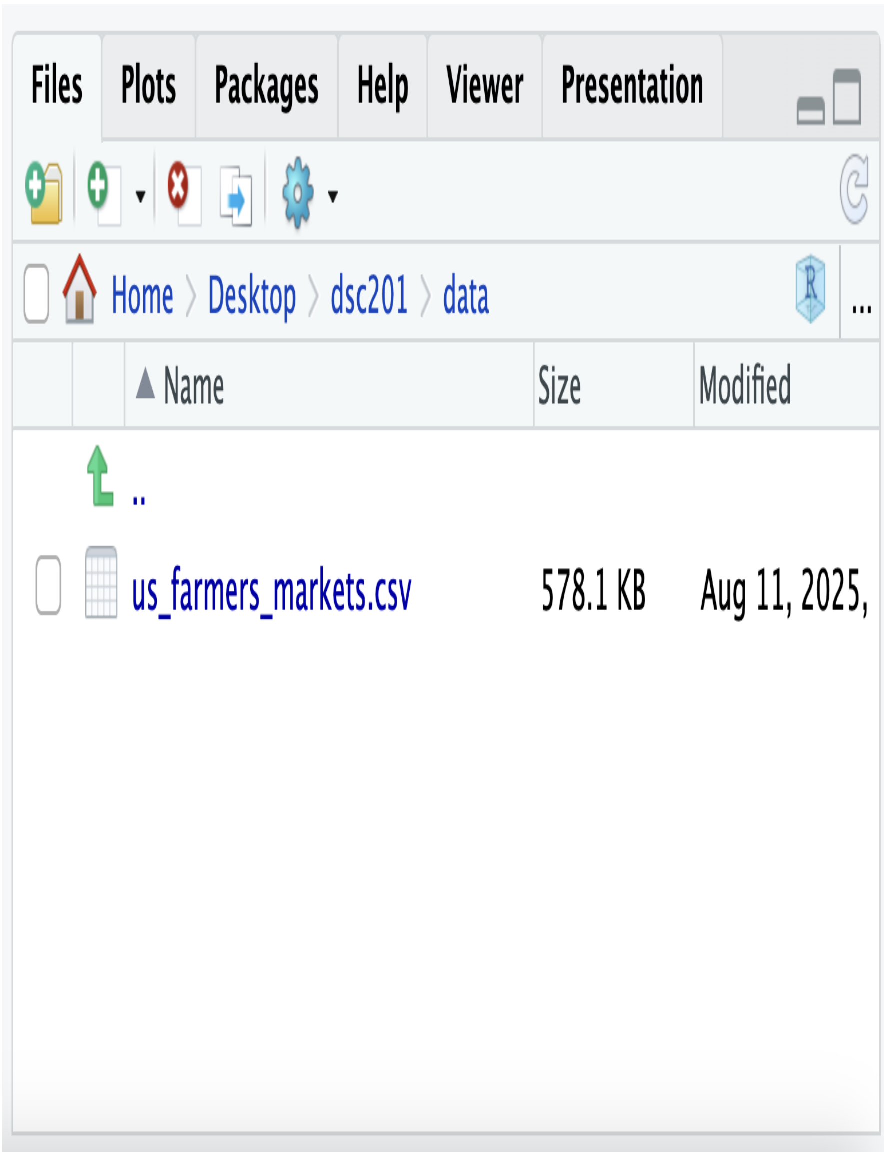
9. In **Please enter the new folder name**, enter data.
10. Click **OK**.

Important: Name the folder `data` **(all lowercase)**. Using all lowercase avoids issues on operating systems where `Data` and `data` are treated as different folders.

Step 3: Download the `us_farmers_markets.csv` data file

11. Download the `us_farmers_markets.csv` dataset from the course Moodle page.

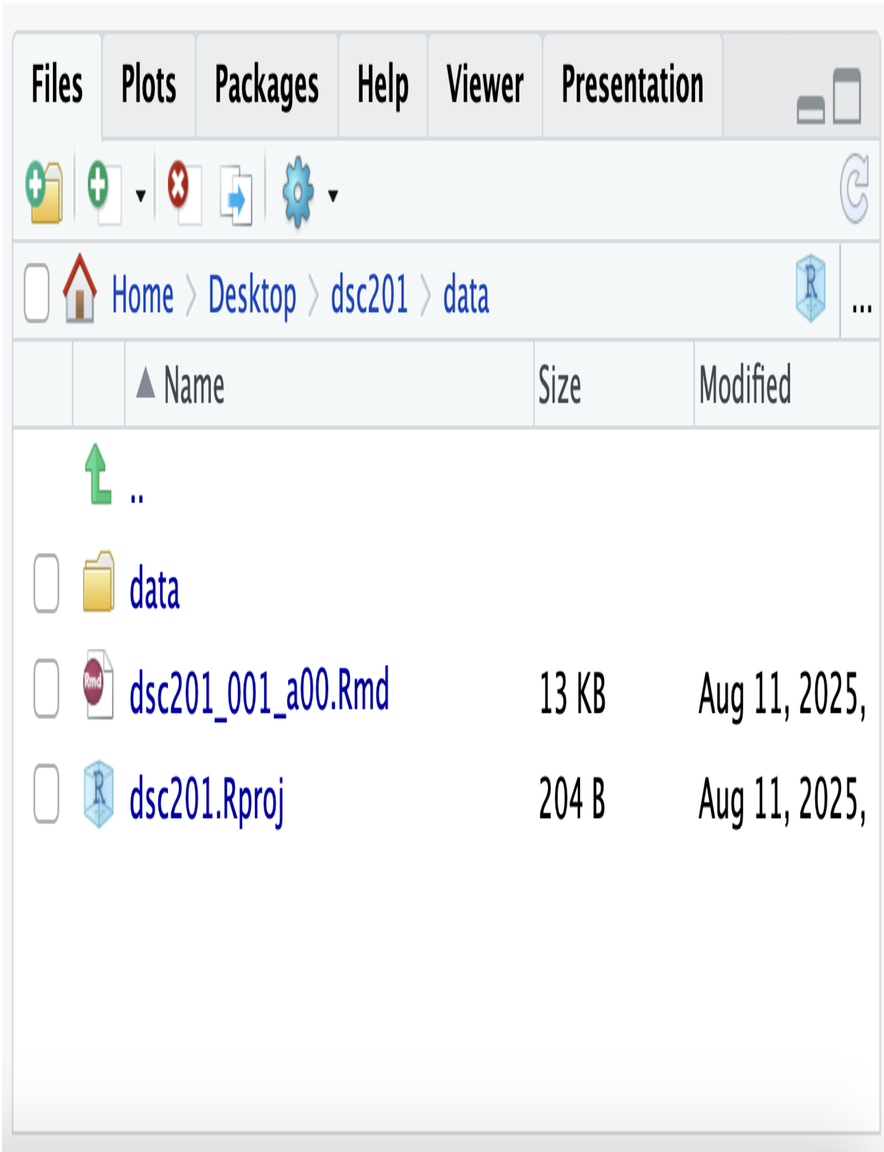
12. Locate it in your browser's **Downloads** folder then move it manually into your `dsc201/data` RProject folder.
13. Confirm the file is in the right place by checking in RStudio's **Files** pane. You should see it in `data/us_farmers_markets.csv`.



Step 4: Download the `.Rmd` file

14. Download the `dsc201_001_a00.Rmd` RMarkdown file from the course Moodle page.

15. Locate it in your browser's **Downloads** folder then move it to your `dsc201` RProject folder (the root folder, not inside `data`).
16. Confirm the file is in the right place by checking in RStudio's **Files** pane. You should see it in `dsc201/ds201_001_a00.Rmd`.



Step 5: Always open the `.Rproj` file

17. In your computer's file explorer, double-click `dsc201.Rproj` to start each session. This ensures your working directory and relative paths work correctly.

Knitting to PDF

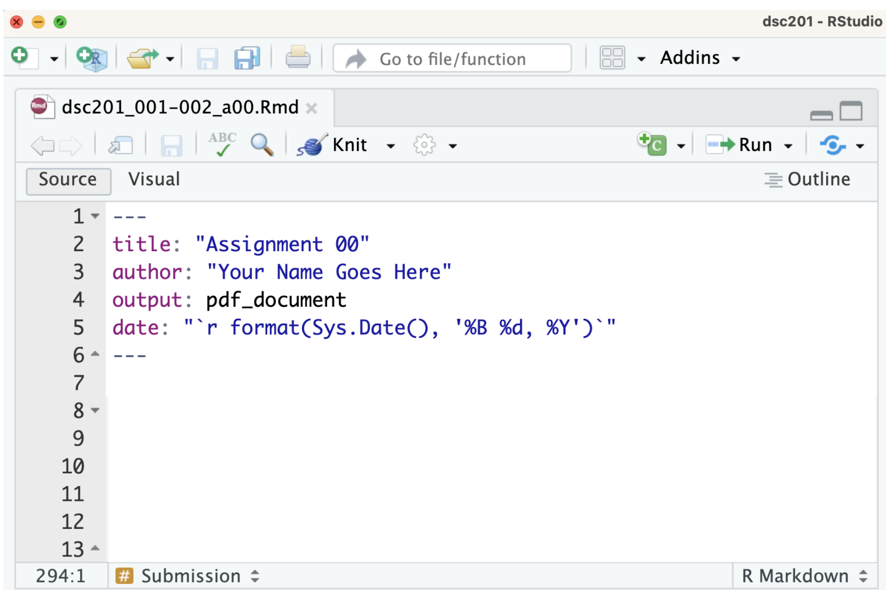
Knitting Assignment 00

In RStudio, **knitting** is the process of taking a source document written in R Markdown (`.Rmd`) and combining the text, code, and code output into a single rendered file, such as HTML, PDF, or Word. When a document is knitted, RStudio runs all code chunks, captures their output (e.g., tables, plots, and summaries), and integrates the results with the written text.

Knitting to PDF

Using the YAML Header

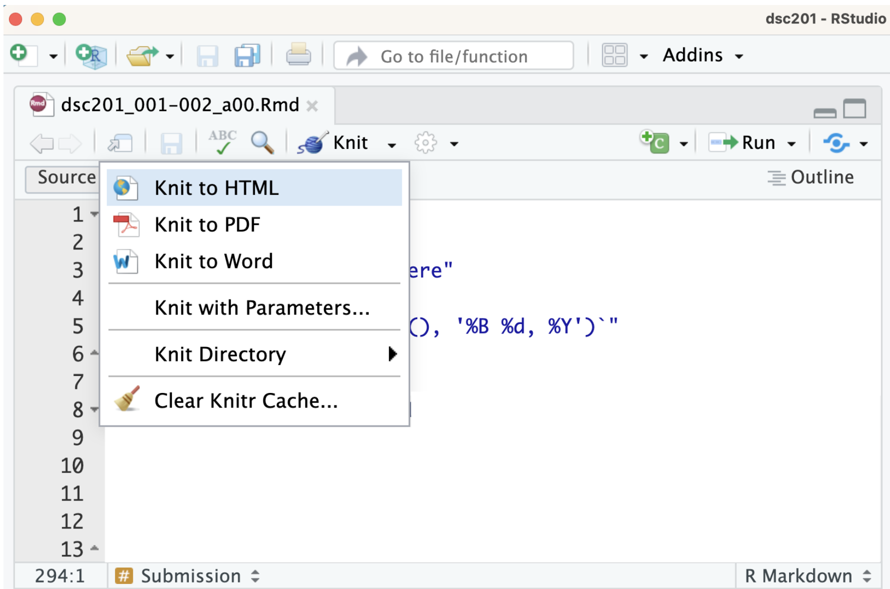
To knit directly to PDF by default, specify PDF output in the YAML header at the top of your document.



Using the Knit Menu

To manually knit a document to PDF:

1. Click the **Knit** button in RStudio.
2. Select **Knit to PDF** from the drop-down menu.



Common Knitting Errors

Missing `knitr`

If you see the following message when attempting to knit

> Required package 'knitr' is missing. Would you like to install it?

the `knitr` package has not yet been installed on your computer.

Click **Yes** and RStudio will automatically install `knitr` and any required dependencies. Once installation is complete, knitting should begin again automatically.

If the document knits successfully, no further action is required.

Missing LaTeX

If you instead see an error similar to the one below:

> **Warning**
>
> Error: LaTeX failed to compile RMarkdown.tex. See https://yihui.org/tinytex/r/#debugging for debugging tips. In addition: Warning message: In system2(..., stdout = if (use_file_stdout()) f1 else FALSE, stderr = f2) : '"pdflatex"' not found
> Execution halted No LaTeX installation detected (LaTeX is required to create PDF output). You should install a LaTeX distribution for your platform: https://www.latex-project.org/get/

your computer does not have a LaTeX distribution installed. To create PDF files from R Markdown, you must install a LaTeX distribution such as TinyTeX.

Proceed to the instructions below.

Installing TinyTeX for PDF Output

To knit R Markdown documents to PDF, your computer needs a LaTeX installation. We recommend TinyTeX because it is lightweight, easy to install, and integrates well with RStudio.

TinyTeX vs. `tinytex`

TinyTeX

TinyTeX is a lightweight LaTeX distribution used to compile R Markdown documents into PDF files. It performs the actual PDF creation process.

`tinytex`

`tinytex` is an R package that installs and manages TinyTeX from within RStudio.

Steps to Set Up TinyTeX

Step 1: Install the `tinytex` Package

- Open RStudio.
- In the **Console**, run:

```r
install.packages("tinytex")
```

Note: This installs the `tinytex` package, which manages the TinyTeX installation.

Step 2: Install TinyTeX

- In the **Console**, run:

```r
tinytex::install_tinytex()
```

Note: This command downloads and installs TinyTeX on your computer and may take several minutes.

Step 3: Verify the Installation

- In the **Console**, run:

```r
tinytex::tinytex_root()
```

If the command returns a file path, TinyTeX has been installed successfully.

Example paths:

- macOS/Linux: `"/usr/local/texlive/..."`
- Windows: `"C:\\Users\\YourName\\AppData\\Roaming\\TinyTeX\\..."`

Step 4: Knit to PDF

- Open your `.Rmd` file in RStudio.
- Verify that the YAML header specifies PDF output:

```yaml
---
title: "Your Title"
author: "Your Name"
output: pdf_document
---
```

- Click **Knit**.
- Select **Knit to PDF** if prompted.

RStudio will now use TinyTeX to generate a PDF version of your document.

Tag System Overview

Purpose

This section explains the tag system used in grading and feedback for activity assignments, programming assignments, and final project milestones.

The goal of this system is to provide transparent, consistent, and instructional feedback that helps you understand how your work was evaluated and how to improve.

Tag System Overview

Assignment comments may include two kinds of tags:

- **Severity tags** describe how serious the issue is and determine the point deduction.
- **Content tags** describe what kind of issue occurred.

Tags appear at the start of comments in this format:

`[SEVERITY][CONTENT-TAG]`

For example:

`[S][CODE-REQ]`

This means the response has a major issue related to the code that was submitted to answer the question.

Severity Tags

Severity tags indicate the overall impact of an issue on correctness or completion. The severity tag determines the point deduction.

Multiple content tags may apply to a response, but only one severity level is assigned.

Content Tags

Content tags identify issues related to the completeness, correctness, clarity, interpretation, organization, or communication of a response.

The specific content tags available differ by assignment type.

Multiple Tags and One Deduction

Multiple tags may apply to the same response. However, deductions reflect the overall severity of the issue rather than treating each tag as a separate deduction.

Content tags identify the specific issues present, while the severity tag determines the point deduction.

For example, consider the following question:

> Calculate the average calories for each restaurant and identify the restaurant with the highest average calories.

A student submits:

```
ms2022 |>
  count(restaurant)
```

Possible tags:

`[S]` `[CODE-REQ]` `[CODE-RESULT]` `[REQ-INCOMP]`.

- `[S]` because most required elements are missing or incorrect.
- `[CODE-REQ]` because the required computation is missing.
- `[CODE-RESULT]` because the result does not support the analysis requested.
- `[REQ-INCOMP]` because the question was not fully answered.

Although multiple content tags apply, the response receives one severity level based on the overall quality of the answer.

Activity Assignments

Grading

Activity assignments are worth 5 points total. Points are awarded based on submission timeliness and submission compliance.

- **1 point** for submission timeliness.
- **4 points** for submission requirements, which include completing all parts of the activity, following formatting requirements, and submitting all required components.

Severity Tags

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
`[T-ON]`	0.0	1.0	Submitted on time
`[T-L1]`	0.625	0.375	Submitted 1 day late
`[T-L2]`	0.775	0.225	Submitted 2 days late
`[T-N]`	1.0	0.0	Submitted more than 2 days late or not submitted

Submission Compliance

Severity Tag	Points Deducted	Points Earned	Explanation
`[C-M]`	0.0	4.0	Submission requirements met
`[C-C]`	0.5	3.5	Required component missing
`[C-S]`	2.0	2.0	Required section missing
`[C-F]`	3.0	1.0	Incorrect submission format
`[C-N]`	4.0	0.0	Not submitted

Content Tags

Assignment Compliance

Tag	Explanation
`[REQ-INCOMP]`	Required part of the question was not answered.
`[REQ-REF]`	Required reference or citation is missing.
`[REQ-MISS]`	Required element is missing.

Writing & Communication

Tag	Explanation
`[WR-FLOW]`	Response lists disconnected observations instead of explaining ideas in sentences.
`[WR-FMT]`	Excessive bolding, unnecessary headings, or inconsistent spacing distracts from readability.

Programming Assignments

Grading

Programming assignments are worth 15 points total. Points are awarded based on participation and completion of the assignment questions.

Participation [5 Points]

- **4 points** for submission timeliness.
- **1 point** for submission compliance, which includes following formatting requirements and submitting all required components.

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
`[T-ON]`	0.0	4.0	Submitted on time
`[T-L1]`	2.0	2.0	Submitted 1 day late
`[T-L2]`	3.0	1.0	Submitted 2 days late
`[T-N]`	5.0	0.0	Submitted more than 2 days late or not submitted

Submission Compliance

Severity Tag	Points Deducted	Points Earned	Explanation
`[C-M]`	0.0	1.0	Submission requirements met
`[C-N]`	0.0625	0.9375	Name missing or incorrect
`[C-D]`	0.0625	0.9375	Date missing or incorrect
`[C-S]`	0.0625	0.9375	Section missing
`[C-F]`	1.0	0.0	Incorrect submission format

Severity Tag	Points Deducted	Points Earned	Explanation

Questions [10 Points]

- **1 point** per question for a total of 10 questions.

Programming assignments include code-based questions and may also include written-response questions. Unless otherwise specified, written responses must:

- be written in third person,
- use proper grammar, punctuation, and spelling, and
- be formatted using standard font (e.g., no excessive bolding or italics).

Severity Tags

Severity Tag	Points Deducted	Points Earned	Explanation
`[C]`	0.0	1.0	The response correctly addresses the question and meets all requirements.
`[e]`	0.0625	0.9375	The work is correct, with issues limited to presentation or precision.
`[m]`	0.125	0.875	The core idea is correct, but some details are missing or unclear.
`[M]`	0.25	0.75	A key element is missing or incorrect.
`[MJ]`	0.50	0.50	The main idea or a required element is missing or incorrect.
`[S]`	0.75	0.25	An attempt was made, but most elements are missing or incorrect.
`[X]`	1.00	0.00	Missing or required work is not present.

Content Tags

Assignment Compliance

Tag	Explanation
`[REQ-INC]`	Incorrect assignment submitted.

Tag	Explanation
[REQ-FMT]	Notebook or file structure was altered, or an incorrect format was submitted.

Question Compliance

Tag	Explanation
[REQ-INCOMP]	Required part of the question was not answered.
[REQ-REF]	Required reference or citation is missing.
[REQ-VIS]	The visualization requirement was not met or properly followed.

Code & Output

Tag	Explanation
[CODE-ERR]	Code produces a syntax or runtime error and does not execute successfully.
[CODE-PH]	Placeholder or instructional comment was not removed (e.g., `# YOUR CODE GOES HERE`).
[CODE-FMT]	Comments and/or code run off the page or are not formatted for readability.
[CODE-STYLE]	Code works but does not follow required style or structure.
[CODE-OUT]	Entire datasets or unnecessary output are displayed when only summary output was needed.
[CODE-MISS]	Required output is missing, hidden, truncated, or difficult to read.
[CODE-REQ]	Required code, computation, or logic is missing.
[CODE-RESULT]	Code runs without errors, but the output does not answer the question or support the analysis.
[CODE-EXTRA]	Extra code is included that does not contribute to the analysis.

Visualizations

Tag	Explanation
[VIS-CLAR]	Visualization is cluttered, overcrowded, or difficult to interpret visually.
[VIS-HIST]	Histogram construction misrepresents continuous data.

Tag	Explanation
`[VIS-CHOICE]`	The visualization type is not appropriate for the structure or type of data being displayed (e.g., using a scatterplot or box plot for discrete categorical data, or using a bar chart for continuous numerical data).
`[VIS-LAB]`	The visualization is not labeled clearly or appropriately (e.g., axis labels, titles).

Writing & Communication

Tag	Explanation
`[WR-3P]`	Uses first-person language instead of third-person (e.g., "I think the graph shows…").
`[WR-OBJ]`	Uses subjective or informal language instead of objective description (e.g., "This graph is really good and interesting.").
`[WR-FLOW]`	Response lists disconnected observations instead of explaining ideas in sentences or paragraphs.
`[WR-FMT]`	Excessive bolding, unnecessary headings, or inconsistent spacing distracts from readability.
`[WR-PH]`	Instructional text such as "Replace this text with your answer" was not removed.
`[WR-REQ]`	A required written response is missing.
`[WR-CODE]`	References commands or functions in the written interpretation (e.g., "I used `group_by()` to calculate the averages.").
`[WR-QNUM]`	Includes question numbers in the written response (e.g., "Question 3:" included directly in the written interpretation).

Interpretation & Reasoning

Tag	Explanation
`[INT-REQ]`	A required explanation or interpretation component is missing.
`[INT-DESC]`	Output is reported but not interpreted (e.g., "The mean age is 42.3." with no explanation of what the value means in context).
`[INT-VAL]`	Interpretation lacks specific numerical values (e.g., "One group was higher than the other" without reporting numerical values).
`[INT-CTX]`	Results are not explained in context or why they matter.
`[INT-REL]`	Variables are discussed without explaining their relationship (e.g., mentions two variables separately without explaining how they relate).

Tag	Explanation
[INT-INF]	Interprets information that is not shown in the output (e.g., claims a trend or relationship that is not supported by the displayed output).
[INT-COR]	Uses correlation without verifying that it is an appropriate measure of the relationship.
[INT-CA]	Uses causal language such as *affects*, *impacts*, or *causes* (e.g., “More exercise causes higher happiness levels”) based only on exploratory analysis.

Final Project Milestones

Final Project Milestones

Milestone 1

In this milestone, you will identify a dataset, demonstrate that it satisfies the project requirements, and create a datasheet documenting the variables, structure, and context of the dataset.

Dataset Approval Request

The Dataset Approval Request is worth 15 points total. Points are awarded based on submission timeliness and dataset approval.

Detailed submission instructions are available in Moodle.

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
`[T-ON]`	0.0	5.0	Submitted on time
`[T-L1]`	2.0	3.0	Submitted 1 day late
`[T-L2]`	3.5	1.5	Submitted 2 days late
`[T-L3]`	4.0	1.0	Submitted more than 2 days late
`[T-N]`	5.0	0.0	Not submitted

Dataset Approval

Severity Tag	Points Deducted	Points Earned	Explanation
`[C]`	0.0	10.0	The dataset meets all requirements and all required submission elements are present. **Dataset approved.**

Severity Tag	Points Deducted	Points Earned	Explanation
[e]	0.75	9.25	One or more minor issues must be addressed before the dataset can be approved. **Dataset not yet approved.**
[m]	2.0	8.0	Some requirements are unmet, unclear, or insufficiently documented. Revision is required before the dataset can be approved. **Dataset not yet approved.**
[M]	5.0	5.0	Several requirements are unmet or unclear. Significant revision or selection of an alternate dataset may be required. **Dataset not yet approved.**
[S]	8.0	2.0	Many requirements are unmet or violated. The dataset is not viable in its current form and should be replaced. **Dataset not yet approved.**
[X]	10.0	0.0	Missing or required work is not present. **Dataset not yet approved.**

Content Tags

Dataset Structure & Size Issues

Tag	Explanation
[D-FTYPE]	File type issue.
[D-FSIZE]	File size issue.
[D-RANGE]	The dataset does not meet the 500–10,000 row requirement.
[D-VAR]	The dataset does not include at least 10 variables.
[D-CAT]	Fewer than 2 categorical variables are present.
[D-NUM]	Fewer than 2 numerical variables are present.

Dataset Suitability Issues

Tag	Explanation
[D-TS]	The dataset is primarily time-series data.
[D-REP]	The dataset contains repeated measurements of the same subject.
[D-PII]	Personally identifiable information (PII) is present in the dataset.

Tag	Explanation
[D-OVR]	The dataset is overly common or has extensive publicly available analyses.
[D-AN]	The dataset is unlikely to support meaningful analysis using the methods covered in this course.

Dataset Access & Documentation Issues

Tag	Explanation
[D-UO]	The unit of observation is unclear or incorrectly described.
[D-LINK]	The dataset link is missing, broken, or requires a personal login.

Dataset Justification & Research Question Issues

Tag	Explanation
[J-FIT]	The justification does not clearly explain why the dataset fits the project goals.
[J-REQ]	The justification does not explain how the dataset meets the project requirements.
[J-RQ]	Example research questions are missing or unclear.

Datasheet

The Datasheet is worth 15 points total. Points are awarded based on submission timeliness and dataset approval.

Detailed submission instructions are available in Moodle.

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
[T-ON]	0.0	5.0	Submitted on time
[T-L1]	2.0	3.0	Submitted 1 day late
[T-L2]	3.5	1.5	Submitted 2 days late
[T-L3]	4.0	1.0	Submitted more than 2 days late
[T-N]	5.0	0.0	Not submitted

Datasheet

Severity Tag	Points Deducted	Points Earned	Explanation
[C]	0.0	10.0	The datasheet fully meets all requirements, addresses all specified criteria, and is well organized and clearly written.
[e]	0.75	9.25	The datasheet is complete and clear, with only minor issues related to presentation, organization, or precision.
[m]	2.0	8.0	The datasheet is generally complete and understandable, but some information is missing, unclear, or insufficiently documented.
[M]	5.0	5.0	Several required elements are missing, incomplete, or unclear. Meaningful revision is required.
[S]	8.0	2.0	Many required elements are missing or unclear; the datasheet does not meaningfully document the dataset.
[X]	10.0	0.0	Missing or required work is not present.

Content Tags

Datasheet Completeness Issues

Tag	Explanation
[DS-MISS]	One or more variables are missing required documentation elements.
[DS-PART]	Variable descriptions are present but lack sufficient detail or clarity.
[DS-SRC]	Dataset source is missing or unclear.

Datasheet Variable Documentation Issues

Tag	Explanation
[DS-FNAME]	The filename does not adhere to snake_case.
[DS-VNAME]	One or more variable names do not adhere to snake_case.
[DS-DESC]	A variable description does not clearly explain the meaning or context of the variable.
[DS-TYPE]	The data type used to store a variable is missing or incorrect.
[DS-UNIT]	The unit of measurement is missing or unclear where one is needed.

Tag	Explanation
[DS-VAL]	Possible values, categories, or coding schemes are missing or unclear.

Datasheet Organization & Clarity Issues

Tag	Explanation
[DS-ORG]	The organization of the datasheet makes it difficult to follow.
[DS-WRITE]	Writing is unclear, imprecise, or difficult to interpret.
[DS-CONTEXT]	The datasheet does not provide sufficient context to understand the dataset and its variables.
[DS-FMT]	Formatting distracts from clarity (e.g., unnecessary bolding, headings, or spacing).
[DS-PH]	Placeholder or instructional text was not removed.

Datasheet Submission Format Issues

Tag	Explanation
[REQ-PDF]	The datasheet was not submitted as a PDF.
[REQ-UP]	The file was not uploaded correctly to Moodle.

Milestone 2

Data Dive Using R and Python

In these milestones, you will use R and Python to explore your dataset, identify patterns and potential issues, and develop insights that will support your final project analysis.

See Programming Assignments for grading criteria.

Milestone 3

In this milestone, you will develop research questions, create an analysis plan, and set up the notebook that will serve as the foundation for your final project analysis.

Research Questions

The Research Questions milestone is worth 15 points total. Points are awarded based on submission timeliness and the quality of the research questions.

Detailed submission instructions are available in Moodle.

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
[T-ON]	0.0	5.0	Submitted on time
[T-L1]	2.0	3.0	Submitted 1 day late
[T-L2]	3.5	1.5	Submitted 2 days late
[T-L3]	4.0	1.0	Submitted more than 2 days late
[T-N]	5.0	0.0	Not submitted

Research Questions

Severity Tag	Points Deducted	Points Earned	Explanation
[C]	0.0	10.0	Three clear, specific, and focused questions are provided. Each question involves two or more variables, avoids causal language, and includes a brief justification.
[e]	0.75	9.25	The research questions are strong overall, with only minor issues related to wording, specificity, variable identification, or explanation.
[m]	2.0	8.0	The research questions are generally appropriate, but one or more questions are unclear, insufficiently specific, weakly justified, or do not fully meet the stated requirements.
[M]	5.0	5.0	Several research questions are unclear, weakly justified, overly broad, or written as yes/no questions. Significant revision is required.
[S]	8.0	2.0	Most research questions do not meet the requirements or examine meaningful relationships between variables.
[X]	10.0	0.0	Missing or required work is not present.

Content Tags

Research Question Completeness Issues

Tag	Explanation
[RQ-NUM]	Fewer than three research questions are provided.
[RQ-LIST]	Variables are referenced but not specifically listed.
[RQ-VAR]	One or more questions do not involve at least two variables.
[RQ-CREATE]	Created variables are not clearly defined or explained.

Research Question Clarity & Specificity Issues

Tag	Explanation
[RQ-CLR]	One or more research questions are unclear or difficult to interpret.
[RQ-SPC]	Questions are too broad, vague, or lack a clear analytical focus.
[RQ-YN]	One or more questions can be answered with a simple yes or no.

Research Question Variable Selection & Relationship Issues

Tag	Explanation
[RQ-VSEL]	Variable choices are not well justified or are poorly matched to the questions.
[RQ-REL]	Questions do not clearly explore relationships between variables.
[RQ-EDA]	Questions are not well suited for exploratory data analysis.

Research Question Causality Issues

Tag	Explanation
[RQ-CAUSE]	Research questions use causal language or imply causal relationships.

Research Question Submission & Formatting Issues

Tag	Explanation
[REQ-PDF]	Submission was not entered in the text box or uploaded as a PDF.
[REQ-FMT]	Formatting makes the submission difficult to read or interpret.

Analysis Notebook Setup

The Analysis Notebook Setup is worth 15 points total. Points are awarded based on submission timeliness and completion of the notebook setup.

Detailed submission instructions are available in Moodle.

What to Include

Your notebook should contain the following sections:

- An introduction
- A dataset description
- Three research questions
- Placeholders for code, visualizations, and discussion for each research question

Example

See: Analysis Plan Notebook Setup Example

The example includes:

- A completed Introduction
- A completed Dataset Description
- One completed Research Question section

Use the example to understand the expected level of detail and organization for your notebook.

Notebook Structure

The notebook template contains the following sections:

- Introduction
- Dataset Description
- Research Question 1
- Research Question 2
- Research Question 3

- Conclusions
- Course Reflection

For Milestone 3, complete the Introduction, Dataset Description, Research Questions, Variables Used, and Planned Analysis sections. Leave the Code, Visualization, Discussion, Conclusions, and Course Reflection sections empty.

Submission Timeliness

Severity Tag	Points Deducted	Points Earned	Explanation
`[T-ON]`	0.0	5.0	Submitted on time
`[T-L1]`	2.0	3.0	Submitted 1 day late
`[T-L2]`	3.5	1.5	Submitted 2 days late
`[T-L3]`	4.0	1.0	Submitted more than 2 days late
`[T-N]`	5.0	0.0	Not submitted

Analysis Notebook Setup

Severity Tag	Points Deducted	Points Earned	Explanation
`[C]`	0.0	10.0	All required sections are completed. Guiding text has been replaced with clear, original content, and no code is included.
`[e]`	0.75	9.25	The notebook is complete overall, with only minor issues related to organization, clarity, or remaining guiding text.
`[m]`	2.0	8.0	Most required sections are completed, but one or more sections are incomplete, unclear, or contain substantial guiding text.
`[M]`	5.0	5.0	Several required sections are incomplete, unclear, or contain placeholder text. Significant revision is required.
`[S]`	8.0	2.0	Many required sections are missing, unchanged from the template, or lack sufficient content to support the final project.
`[X]`	10.0	0.0	Missing or required work is not present.

Content Tags

Tag	Explanation
[NB-GUIDE]	Guiding or placeholder text was not removed from one or more sections.
[NB-MISS]	One or more required sections are missing or empty.
[NB-CLR]	Written responses are unclear, incomplete, or difficult to interpret.
[NB-CODE]	Code was added despite instructions not to include code.
[NB-STRUCT]	Required notebook sections were removed, renamed, or reorganized.

Milestone 4

In this milestone, you will complete your final project notebook using either R or Python. You will conduct your analysis, create visualizations, and communicate your findings by answering the research questions developed in the previous milestones.

Data Analysis & Storytelling

The Data Analysis & Storytelling Notebook is worth 30 points total. Points are awarded based on submission timeliness and completion of the notebook.

Detailed submission instructions are available in Moodle.

Severity Scale for 3-Point Components

Severity Tag	Points Deducted	Points Earned	Explanation
[C]	0.00	3.00	The response correctly addresses the question and meets all requirements.
[e]	0.1875	2.8125	The work is correct, with issues limited to presentation or precision.
[m]	0.375	2.625	The core idea is correct, but some details are missing or unclear.
[M]	0.75	2.25	A key element is missing or incorrect.
[MJ]	1.50	1.50	The main idea or a required element is missing or incorrect.
[S]	2.25	0.75	An attempt was made, but most elements are missing or incorrect.
[X]	3.00	0.00	Missing or required work is not present.

Severity Scale for 2-Point Components

Severity Tag	Points Deducted	Points Earned	Explanation
[C]	0.00	2.00	The response correctly addresses the question and meets all requirements.
[e]	0.125	1.875	The work is correct, with issues limited to presentation or precision.
[m]	0.25	1.75	The core idea is correct, but some details are missing or unclear.
[M]	0.50	1.50	A key element is missing or incorrect.
[MJ]	1.00	1.00	The main idea or a required element is missing or incorrect.
[S]	1.50	0.50	An attempt was made, but most elements are missing or incorrect.
[X]	2.00	0.00	Missing or required work is not present.

Content Tags

The Data Analysis & Storytelling assignment follows the same tags as programming assignments. See Programming Assignments for details.

Programming Style Guide

Purpose

The purpose for this guide is to help you write clear, readable, and professional code from the start. It draws on recommendations from Pruim et al. (2023), Rossum et al. (2001), and Wickham (2015).

General Guidelines

Comments

- Comments start with `#`
- Comments should be sentence case

```
# Load the data
```

- Comments should usually be a short phrase or single sentence that describes the purpose of a block of code. Use additional sentences only to explain non-obvious decisions (e.g., `# Keep only data from 2024. Earlier years use a different reporting# system and are not directly comparable.`)

Pick clear, descriptive names and a standard naming convention

- Use meaningful names (e.g., `average_score`, not `x1`)
- Use `snake_case` for variable and user-defined function names
- Avoid using dots in names (e.g., `average.score`)

Write code in logical blocks

- Use blank lines between steps

```
# Laod the data
code

# Clean the data
code
code
code
```

Use consistent spacing

For assignment statements and arithmetic expressions.

Language	Good Example	Bad Example
R	`height <- feet * 12 + inches`	`height<-feet*12+inches`
Python	`df["lbs"] = df["kg"] / 2.204`	`df["lbs"]=df["kg"]/2.204`

For assigning default values or passing keyword arguements.

Language	Good Example	Bad Example
R	`mean(x, na.rm=TRUE)`	`mean(x, na.rm = TRUE)`
Python	`round(number, ndigits=2)`	`round(number, ndigits = 2)`

Handle long lines thoughtfully

- Keep lines 79 chars
- Use parentheses for long expressions

```
df |>
  summarise(
    Sepal.Length = mean(Sepal.Length),
    Sepal.Width = mean(Sepal.Width),
    .by = Species
  )
```

```python
results = pd.concat(
    [df1, df2, df3],
    ignore_index=True,
    sort=False
)
```

Import libraries and load packages at the beginning

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```r
library(tidyverse)
library(ggplot2)
```

Python-Specific Style

Chain methods when appropriate

```python
df.groupby("category").mean()
```

Define functions clearly

```python
def calculate_mean(values):
    """Return the mean of a list of numbers."""
    return sum(values) / len(values)
```

R-Specific Style

Use the Pipe |>

```r
data |>
  filter(score > 80) |>
  summarise(avg = mean(score))
```

Format pipes with whitespace

- Pipes go at end of line
- Indent by 2 spaces
- Short pipes go one a single line `iris |> summary()`

Format `ggplot2` code

```
ggplot(iris, aes(x = Sepal.Width, y = Sepal.Length)) +
  geom_point() +
  labs(
    x = "Sepal width (cm)",
    y = "Sepal length (cm)",
    title = "Sepal Dimensions by Species"
  )
```

Use `<-` for assignment

```
x <- 10
```

Formatting if statements

```
score <- 10
if (score > 15) {
  print("High")
} else if (score < 15) {
  print("Medium")
} else {
  print("Low")
}
```

Milestone Templates

Milestone 1: Finding a Dataset

Dataset Approval Request Template

> **Note**
>
> A copy of this template can be found here. Create your own copy and complete it before submitting.

Name:

Section:

Course: Introduction to R and Python for Data Science

Date:

Dataset Description

Provide a brief description of the dataset.

Example

> MenuStat is a free nutrition database of foods and beverages served by the nation's top grossing chain restaurants.

Dataset Name: The name of the CSV file.

Example

> `ms_annual_data_2022.xls`

Link: Provide a direct link to the website where the dataset can be accessed or downloaded. The link should work without requiring a personal login whenever possible.

Example

> https://www.menustat.org/data.html

Dataset Dimensions

Number of observations (rows):

Number of variables (columns):

Unit of Observation

Describe what each row in the dataset represents.

Example

> Each row represents a single menu item offered by a restaurant.

Numerical Variable Names and Descriptions

Name	Description
`matched_2021`	Indicator for whether the menu item was also present in the 2021 dataset
`new_item_2022`	Indicator for whether the menu item is new in the 2022 dataset
`serving_size`	Serving size amount
`calories`	Energy content (kcal)
`total_fat`	Total fat (g)
`saturated_fat`	Saturated fat (g)
`trans_fat`	Trans fat (g)
`cholesterol`	Cholesterol (mg)
`sodium`	Sodium (mg)
`carbohydrates`	Total carbohydrates (g)
`dietary_fiber`	Dietary fiber (g)
`sugar`	Total sugar (g)
`protein`	Protein (g)
`potassium`	Potassium (mg)

Categorical Variable Names and Descriptions

Name.	Description
`menu_item_id`	Unique identifier for the menu item
`food_category`	Category of food or beverage item
`restaurant`	Name of the restaurant chain
`item_name`	Name of the menu item
`item_description`	Description of the menu item
`serving_size_text`	Serving size as reported by the restaurant

Name.	Description
`serving_size_unit`	Unit of measurement for the serving size
`serving_size_household`	Household serving size description

Potential Research Questions and Descriptions

List 2–3 research questions that could be investigated using this dataset.

1.
2.
3.

Example

1. How do calorie counts vary across different food categories?
2. What is the relationship between serving size and calorie content?
3. Which restaurant chains offer menu items with the highest average calorie content?

Justification

Explain why you are interested in this dataset and why it is appropriate for the project.

Datasheet Template

> **Note**
>
> A copy of this template can be found here. Create your own copy and complete it before submitting.

Name

The name of the CSV file

Example

```
ms_annual_data_2022.csv
```

Description

Instructions: Write a brief description of your dataset (3-5 sentences). Your description should:

- Explain what the dataset is about
- Identify the data source
- Summarize the types of information included
- Clearly state the unit of observation (what one row represents)

Example

> MenuStat is a nutrition database containing menu items from large U.S. chain restaurants. The data are collected from nutrition information published on restaurant websites and are maintained by the Harvard Pilgrim Health Care Institute. The dataset includes information about menu items, restaurant names, food categories, serving sizes, and nutritional measures such as calories, fat, sodium, carbohydrates, and protein. Each row represents a single menu item offered by a restaurant.

Source

Instructions: Provide the name of the source and a direct link to the webpage or repository where the dataset was obtained. The link should allow someone else to locate the same dataset.

Example

> **Source:** The Harvard Pilgrim Health Care Institute **Link:** https://www.menustat.org/data.html

Variables

Instructions: Complete the variables table for your dataset. Each row in the table should describe one variable (column) in the dataset.

For each variable, provide the following:

- **Variable name:** The exact column name as it appears in the dataset.
- **Description:** A brief explanation of what the variable represents.
- **Type:** Indicate whether the variable is Cat. (categorical) or Num. (numerical).
- **Units:** The units of measurement for the variable (e.g., students, dollars, years). If the variable is categorical, write N/A.

Example

Name	Type	Description
matched_2021	Cat.	Indicator for whether the menu item was also present in the 2021 MenuStat dataset (1 = yes, 0 = no)
new_item_2022	Cat.	Indicator for whether the menu item is new in the 2022 dataset (1 = yes, 0 = no)
menu_item_id	Cat.	Unique identifier for the menu item
food_category	Cat.	Category of food or beverage item
restaurant	Cat.	Name of the restaurant chain
item_name	Cat.	Name of the menu item
item_description	Cat.	Description of the menu item
serving_size	Num.	Serving size amount
serving_size_text	Cat.	Serving size as reported by the restaurant
serving_size_unit	Cat.	Unit of measurement for the serving size (e.g., oz, g, mL)
serving_size_household	Cat.	Household serving size description (e.g., 1 sandwich, 1 slice)
calories	Num.	Energy content (kilocalories)
total_fat	Num.	Total fat (grams)
saturated_fat	Num.	Saturated fat (grams)
trans_fat	Num.	Trans fat (grams)
cholesterol	Num.	Cholesterol (milligrams)
sodium	Num.	Sodium (milligrams)
carbohydrates	Num.	Total carbohydrates (grams)
dietary_fiber	Num.	Dietary fiber (grams)
sugar	Num.	Total sugar (grams)
protein	Num.	Protein (grams)
potassium	Num.	Potassium (milligrams)
notes	Cat.	Additional notes provided by MenuStat
calories_text	Cat.	Calories reported as text by the restaurant
total_fat_text	Cat.	Total fat reported as text by the restaurant
saturated_fat_text	Cat.	Saturated fat reported as text by the restaurant
trans_fat_text	Cat.	Trans fat reported as text by the restaurant
cholesterol_text	Cat.	Cholesterol reported as text by the restaurant
sodium_text	Cat.	Sodium reported as text by the restaurant
carbohydrates_text	Cat.	Carbohydrates reported as text by the restaurant
dietary_fiber_text	Cat.	Dietary fiber reported as text by the restaurant
sugar_text	Cat.	Sugar reported as text by the restaurant
protein_text	Cat.	Protein reported as text by the restaurant

Notes: The variables notes, calories_text, total_fat_text, saturated_fat_text, trans_fat_text, cholesterol_text, sodium_text, carbohydrates_text, `dietary_fiber_text`, `sugar_text`, and `protein_text`contain only missing values in the 2022 dataset.

Milestone 2: Data Dive

Data Dive Using R

Data Dive Using Python

Milestone 3: Analysis Plan

Developing Research Questions

The purpose of this guide is to help you develop effective research questions for your final project. Strong research questions focus your exploratory data analysis and help identify patterns, trends, and relationships worth investigating.

What to Include in Your Submission

- Three research questions, each involving two or more variables.

- A brief explanation of why you chose these variables and why the relationship is worth exploring.

General Guidelines

- Focus on exploring relationships, patterns, and trends in the data

- Ensure questions are clear, specific, and focused

- Prioritize comparisons (across groups, categories, or over time)
- Avoid questions that can be answered with a simple *yes* or *no*
- Avoid language that implies causation (e.g., *affect*, *impact*)

Guidance for Writing Research Questions

Use Two or More Variables

Each question must examine a relationship between variables.

Example

- **Avoid:** *What is the average salary?*
- **Use:** *What is the relationship between years of experience and salary?*

Be Specific and Clear

Clearly state what you are investigating and identify the variables involved.

Example

- **Avoid:** *Is there a connection between studying and grades?*
- **Use:** *How does the number of hours studied relate to final exam scores among high school students?*

Avoid Causal Language

Exploratory analysis identifies relationships—not cause and effect.

Example

- **Avoid:** *Does income affect education level?*

- **Use:** *What is the relationship between income and education level?*

Consider Adding Depth

You can strengthen a research question by examining how a relationship differs across groups or categories.

Example

- **Good:** *What is the relationship between physical activity and heart rate?*
- **Better:** *What is the relation

Analysis Plan Notebook Setup

> **Note**
>
> Download the template needed for this project:
>
> - Analysis Plan Notebook Template (Jupyter Notebook)
> - Analysis Plan Notebook Template (RMarkdwn)
>
> Create your own copy of each template and complete it before submitting.

Introduction

Write a brief introduction describing the purpose of your project and the topic you plan to investigate.

Example

> This project explores trends in restaurant menu items using nutritional information collected from several national restaurant chains. The goal of the analysis is to identify patterns in menu item calories and investigate how calorie counts vary across different restaurants and menu categories. Understanding these patterns may provide insight into differences in restaurant offerings and nutritional content.

Dataset Description

Describe your dataset, including its purpose, unit of observation, size, and source.

Example

> This dataset contains nutritional information for menu items offered by several restaurant chains. The data was collected to provide consumers with information about the nutritional content of restaurant meals. Each row in the dataset represents a single menu item. The dataset contains 9,569 observations. No observations have been removed at this stage of the analysis. The variables most relevant to this project include the restaurant name, menu category, and calorie count. These variables will be used to investigate differences in nutritional content across restaurants and menu item types. **Source:** MenuStat REseteraunt Nutrition Database. **Link:** https://www.menustat.org/uploads/1/4/1/6/141624194/ms_annual_data_2022.xls

Variables

List and describe the variables in your dataset. For each variable, provide its name, type (categorical or numerical), and a brief description of what it represents.

Example

Name	Type	Description
`matched_2021`	Cat.	Indicator for whether the menu item was also present in the 2021 MenuStat dataset (`1` = yes, `0` = no)
`new_item_2022`	Cat.	Indicator for whether the menu item is new in the 2022 dataset (`1` = yes, `0` = no)
`menu_item_id`	Cat.	Unique identifier for the menu item
`food_category`	Cat.	Category of food or beverage item
`restaurant`	Cat.	Name of the restaurant chain
`item_name`	Cat.	Name of the menu item
`item_description`	Cat.	Description of the menu item
`serving_size`	Num.	Serving size amount
`serving_size_text`	Cat.	Serving size as reported by the restaurant
`serving_size_unit`	Cat.	Unit of measurement for the serving size (e.g., oz, g, mL)
`serving_size_household`	Cat.	Household serving size description (e.g., 1 sandwich, 1 slice)
`calories`	Num.	Energy content (kilocalories)
`total_fat`	Num.	Total fat (grams)
`saturated_fat`	Num.	Saturated fat (grams)
`trans_fat`	Num.	Trans fat (grams)
`cholesterol`	Num.	Cholesterol (milligrams)
`sodium`	Num.	Sodium (milligrams)
`carbohydrates`	Num.	Total carbohydrates (grams)
`dietary_fiber`	Num.	Dietary fiber (grams)
`sugar`	Num.	Total sugar (grams)
`protein`	Num.	Protein (grams)
`potassium`	Num.	Potassium (milligrams)

Research Question 1

Question

How does the average calorie content of menu items vary across restaurants?

Variables Used

List the variables that will be used to answer this research question and briefly describe their role in the analysis.

Example

Name	Role in Analysis
`restaurant`	Grouping variable used to compare restaurants
`calories`	Numerical variable used to measure calorie content

Planned Analysis

Example

> To answer this question, I will compare the average calorie content of menu items across restaurants. I will calculate summary statistics, including the mean and median calorie counts for each restaurant. This analysis will help identify whether some restaurants tend to offer higher-calorie menu items than others.

Code

```
# Add code for this analysis here.
```

Conclusions

Summarize the main findings related to the research question. Discuss any patterns, relationships, or unexpected results observed in the analysis.

Research Question 2

Question

> What is the relationship between serving size and calorie content, and how does this relationship differ across food categories?

Variables Used

List the variables that will be used to answer this research question and briefly describe their role in the analysis.

Example

Name	Role in Analysis
`serving_size`	Serving size amount
`calories`	Numerical variable used to measure calorie content
`food_category`	Category of food or beverage item

Planned Analysis

Example

> To answer this question, I will examine the relationship between serving size and calorie content for menu items in the dataset. I will create summary statistics for serving size and calories and investigate whether larger menu items tend to contain more calories. I will then compare this relationship across food categories to determine whether the association between serving size and calorie content differs among different types of menu items.

Code

```
# Add code for this analysis here.
```

Visualization

Describe a visualization that will help answer the research question and explain why it is appropriate.

> I will create a scatterplot with serving size on the x-axis and calorie content on the y-axis. Menu items will be colored by food category to make it easier to compare patterns across categories. This visualization will help identify whether larger serving sizes are associated with higher calorie counts and whether this relationship varies among food categories.

```
# Add code to create the visualization here.
```

Conclusions

Summarize the main findings related to the research question. Discuss any patterns, relationships, or unexpected results observed in the analysis.

Research Question 3

Question

> How does the relationship between calories and sodium content differ across restaurant chains?

Variables Used

List the variables that will be used to answer this research question and briefly describe their role in the analysis.

Example

Name	Role in Analysis
`restaurant`	Grouping variable used to compare restaurants
`sodium`	Sodium (mg)
`calories`	Numerical variable used to measure calorie content

Planned Analysis

Example

> To answer this question, I will examine the relationship between calorie content and sodium content for menu items in the dataset. I will calculate summary statistics for both variables and investigate whether menu items with higher calorie counts also tend to contain more sodium. To compare restaurants, I will focus on the restaurant chains with the largest number of menu items and determine whether the relationship between calories and sodium differs among these restaurants.

Code

```
# Add code for this analysis here.
```

Visualization

Describe a visualization that will help answer the research question and explain why it is appropriate.

> I will create a scatterplot with calorie content on the x-axis and sodium content on the y-axis. Points will represent menu items from the restaurant chains with the largest number of menu items in the dataset, with each restaurant displayed using a different color. This visualization will help identify the overall relationship between calories and sodium content and allow comparisons of this relationship across major restaurant chains.

```
# Add code to create the visualization here.
```

Conclusions

Summarize the main findings related to the research question. Discuss any patterns, relationships, or unexpected results observed in the analysis.

Summary

Provide a brief summary of the project, including the research questions investigated and the key findings from your analysis.

Course Reflection

Reflect on what you learned during the project. Discuss any challenges you encountered, skills you developed, and insights you gained from working with data.

Milestone 4: Data Analysis and Storytelling

References

Pruim, Randall, Maria-Cristiana Gîrjău, and Nicholas J. Horton. 2023. "Fostering Better Coding Practices for Data Scientists." *Harvard Data Science Review* 5 (3). https://doi.org/10.1162/99608f92.97c9f60f.

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